

Influence of rainfall time series sample rate in the improvement of decision-making based on high intensity rainfall events

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ABSTRACT

This study aims to assess whether data acquisition frequency can affect the understanding of high-intensity precipitation events. Time series from a weighing gauge and a weighbridge gauge installed at the same site were analysed, with rainfall intensity data acquisition frequencies of 10 seconds and 5 minutes, respectively. To check the consistency of the data obtained by the two types of rainfall measuring instruments, a correlation analysis was carried out between the two time series.

The results relating to acquisition frequency showed that, for an acquisition frequency of 5 minutes, there are significant losses of information (especially during heavy rainfall events) compared to acquisition frequencies of 10 seconds. The correlation analysis showed that the gauges are measuring correctly and that the data is accurate.

1. Introduction

Rainfall measurement is a quantity characterized in the continuous time domain, being its observation provided by data obtained for discrete acquisition intervals. The data acquisition sampling rate can significantly impact the analysis of extreme events by attenuating maximum rainfall intensities, leading to what generally is referred to as a potential reduction or loss of information about maximum selection intensities. The challenge arises when high-intensity events occur in a shorter period than the chosen time interval. Data collected may not accurately capture the maximum rainfall intensity during a specific observation interval in these cases. Consequently, you will need to complete the information provided for decision-making regarding the occurrence of high-intensity events.

Precipitation measurement [1], predominantly applied to forecast models in a meteorological framework, often relies on rain gauges, the most common the tipping bucket rain gauges, whose measurement characteristics are of an integrative nature of the collected quantity (volume of rain collected during interval times) allowing to estimate the precipitation (the output quantity of interest). We perform data acquisition at time intervals typically of 1–5 minutes [1], considering them adequate for monitoring precipitation and for its application in forecasting.

However, the growing impact of climate changes, especially in urban areas, requires an increase in the sensitivity of these measuring systems to be able to obtain information about events with high rainfall intensities in short periods, which are becoming more relevant to model and forecast critical situations for daily life and to support decision-making.

The comparison of the time series should allow us to discuss the errors of the tipping bucket method, considering the weighing method as a reference and its correlation to the magnitude of precipitation

measurements. In this way, we can show that a higher frequency acquisition for rainfall allows for i) a more accurate representation of its patterns (which is essential for capturing changes in intensity that occur in short periods, especially during heavy rainfall events), ii) capturing the peak intensity and discriminate the changes in the characteristics of the phenomenon, in the case of high-intensity rainfall phenomena, which often have short durations and high variability; iii) increasing the accuracy of flood forecasting models, leading to more timely and reliable warnings; iv) assessing water availability to optimize reservoir operations and plan water supply during extreme events; and v) improving the accuracy of hydrological models, leading to more reliable flow and runoff forecasts [2].

The studies developed [3,4] within the framework of this work allow us to discuss the impact of the instrumentation time resolution and the capability to obtain better time discretization of the observable quantity, considering that it reflects a phenomenon of a dynamic nature, and therefore a direct an impact on the decision-making processes. The particularity of the phenomenon shows the presence of high-intensity rainfall events, the detection of which evidence has significant importance for monitoring, modelling and forecasting. The requirement necessitates developing metrological strategies to observe the appropriate quantity.

This study provides a comprehensive overview of the weather station, highlighting its unique facilities and measurable quantities. The focus of this study is on the analysis of precipitation, one of the measurable quantities at the station. A detailed description of the precipitation measuring equipment used to observe the quantity of interest is provided, along with the associated measurement methods and the respective precipitation intensity time series. Notably, six extreme events are identified in both series, with rainfall intensities exceeding 4mm/h [5]. The results of the correlation analysis conducted on the two series are then presented. Finally, the study concludes with insightful



Fig. 1. LNEC meteorological station.



Fig. 2. Weighing rain gauge.

conclusions and a glimpse into future work.

2. Brief description of the LNEC meteorological station

The main objective of developing a meteorological station at LNEC (Fig. 1) was to provide this entity with an infrastructure suitable for research needs in civil engineering. This would allow it to have traceable reference data on a set of measurable quantities, highlighting precipitation.

The need to have data associated with this quantity was due to two main factors: the availability of precipitation data, which, as a rule, is obtained using tipping bucket rain gauges, with less accuracy than the needs for research purposes, and to reduce the time intervals of acquisition, that are usually too long to detect the extreme events; and the growing interest in observing extreme phenomena with an impact on hydrology and urban hydraulics models.

The LNEC meteorological station allows measurements of



Fig. 3. Wind speed and direction sensor and Vaisala relative humidity and temperature sensor [7,8].

precipitation using a weighing rain gauge (shown in Fig. 2), traceable to SI, a primary gravimetric method [6], and internal reference standards. This is unique reference equipment of this nature in Portugal.

Its measuring chain includes an acquisition system that allows data collection with a frequency of 10 s and has been collecting data since September 2023. This station generates energy autonomy through locally installed solar panels and carries out the acquisition process remotely using wireless communication.

The LNEC meteorological station has instrumentation that allows data to be obtained on other meteorological quantities (Fig. 3), namely, temperature, relative humidity, atmospheric pressure, wind speed and direction, and quantities associated with natural lighting. It should be noted that some of these quantities are considered to influence precipitation observations.

The data obtained allows the development of metrological characterization studies of other types of equipment (namely, the use of two other tipping bucket rain gauges of partner entities placed in the exact location, allowing the comparison of their metrological performance with the LNEC weighing rain gauge as reference), and research studies focused on aspects such as rainwater drainage in the city of Lisbon in conditions of extreme rainfall in short periods, an increasingly frequent phenomenon due to the impact associated with climate changes.

3. Measurement of rainfall

Measurement of rainfall, which has various applications in areas such as hydrology, metrology, and meteorology, is essential in assessing the impact of climate change on society as a parameter applied to interpreting extreme precipitation events [9].

As the impact of climate change increases, experts expect the frequency and intensity of these extreme events to rise, making it essential to characterize these events (where maximum rainfall intensities occur) over a shorter period than the conventionally adopted data acquisition periods of 1–5 minutes [1]. This necessity prompts a comparison to determine whether the data conventionally used in modelling phenomena contains sufficient information to characterize these events accurately.

The RStudio © software was used to display the time series associated with each rainfall measuring instrument, process the data from each time series, sort them by time scale, check and eliminate repeated indices, and check and eliminate 'NA' indices.



Fig. 4. Tipping bucket gauge.

Below are presented the two rainfall measuring instruments used in this study, with different characteristics, measurement methods, data acquisition frequencies, and the associated time series representation. These instruments are installed at the same site.

The rainfall weighing gauge tested is the OTT Pluvio2L (Fig. 2), typically used to measure rainfall automatically and then calculate the intensity and amount of rainfall. The OTT Pluvio2L uses the weighing principle [10]. The manufacturing characteristics of the OTT Pluvio2L weighing gauge are a 400 cm² nominal caption area opening and a resolution for the water accumulation measurement equal to 0.01 mm. The quantity analysed in this study is rainfall intensity, in mm/min, and OTT Pluvio 2L, which uses the moving precipitation growth over the last minute before the sampling interval measurement method described in Refs. [1,10]. In Ref. [10], it is stated that this type of measurement is beneficial for determining the exact intensity of precipitation during extreme events. The data acquisition frequency adopted was 10 seconds, and Fig. 5 shows the respective time series from November 3, 2023 to April 4, 2024.

Tipping bucket gauges allow water collection and guidance to a twin bucket balance, with both parts having an equal weight and reference volume. Every time a bucket is full, the balance changes position within a pivot axis, and the other bucket moves into position to collect water while the first one empties the collected water. In this process, the time between each change (tipping) can be measured, allowing the calculation of the rate of rainfall quantity [1,9]. Tipping-bucket gauges (employ a contact closure (reed switch or relay contact), so each tip produces an electrical impulse as a signal output. This mechanism provides a continuous measurement without manual interaction. Fig. 4 presents the rainfall tipping bucket gauge. The quantity analysed is rainfall intensity, in mm/time interval, to compare with the results obtained with weighing. The data acquisition frequency is 5 minutes, and Fig. 6 shows the respective time series from November 3, 2023 to April 4, 2024.

Figs. 5 and 6 show a similar rainfall profile in both series, finding a correlation higher than 0,9 between the two-time series. In the series associated with the weighing gauge, the maximum rainfall intensity was 2,21 mm/min (these values were obtained by the integration of the six observations made in each minute, with a time interval of 10 s). In the series associated with the tipping bucket gauge, the maximum recorded rainfall intensity was 5 mm/5 min (according to the output resolution of the equipment used). It is noticed that the maximum rainfall intensity values recorded by the two measurement methods occurred with a short delay, which may be caused by the distance of location in the station (Fig. 1).

Also, notice that, regarding the eight extreme events observed, the delay difference can be explained by the fact that, due to high precipitation, the water storage capacity of the weighing gauge was reached (see Fig. 7). When the bucket reaches its storage capacity (when the level in the bucket is equal to or greater than 80 %, which corresponds to 600 mm of rainfall), warnings are displayed on the datalogger associated with the weighing gauge.

[10] describes the procedure for emptying the bucket. After emptying the bucket, the manufacturer's software performs an accuracy test [10], allowing for local assessment of the weighing gauge.

The information from the accuracy test is complemented by the results obtained from the calibration carried out on the weighing gauge in the laboratory [6], making it possible to identify the zero of the weighing gauge that corresponds to the weight of the empty bucket [11].

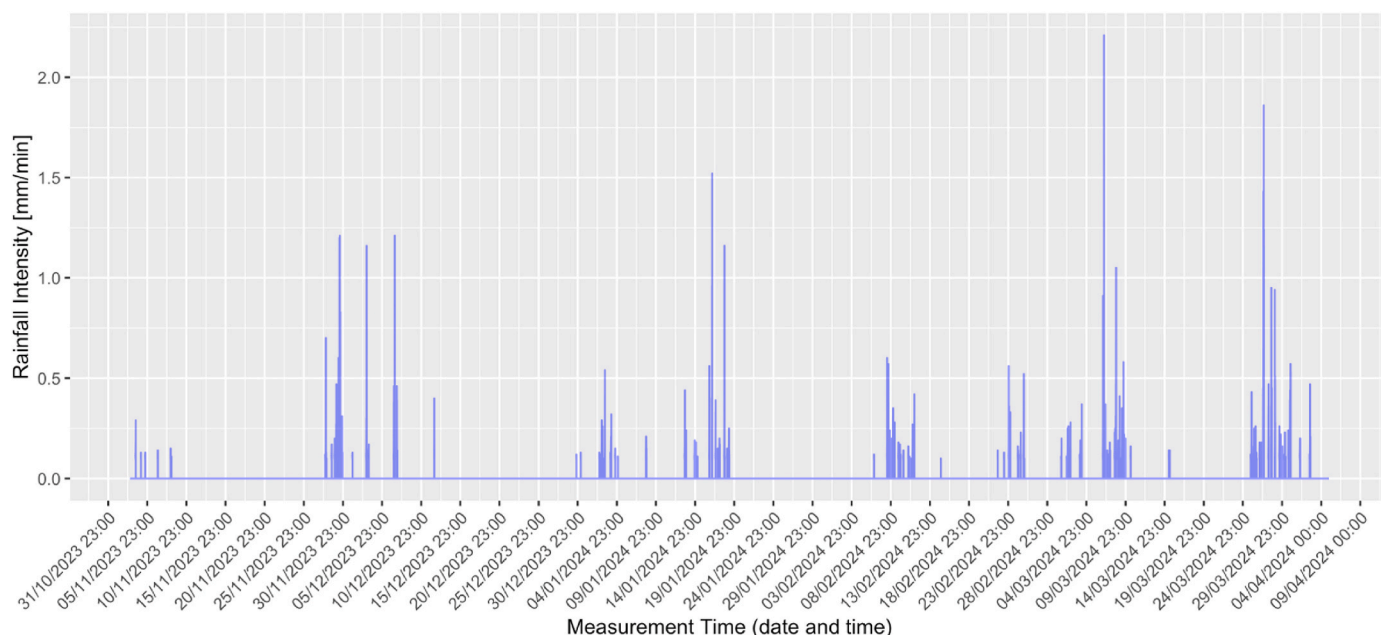


Fig. 5. Time series associated with the weighing gauge.

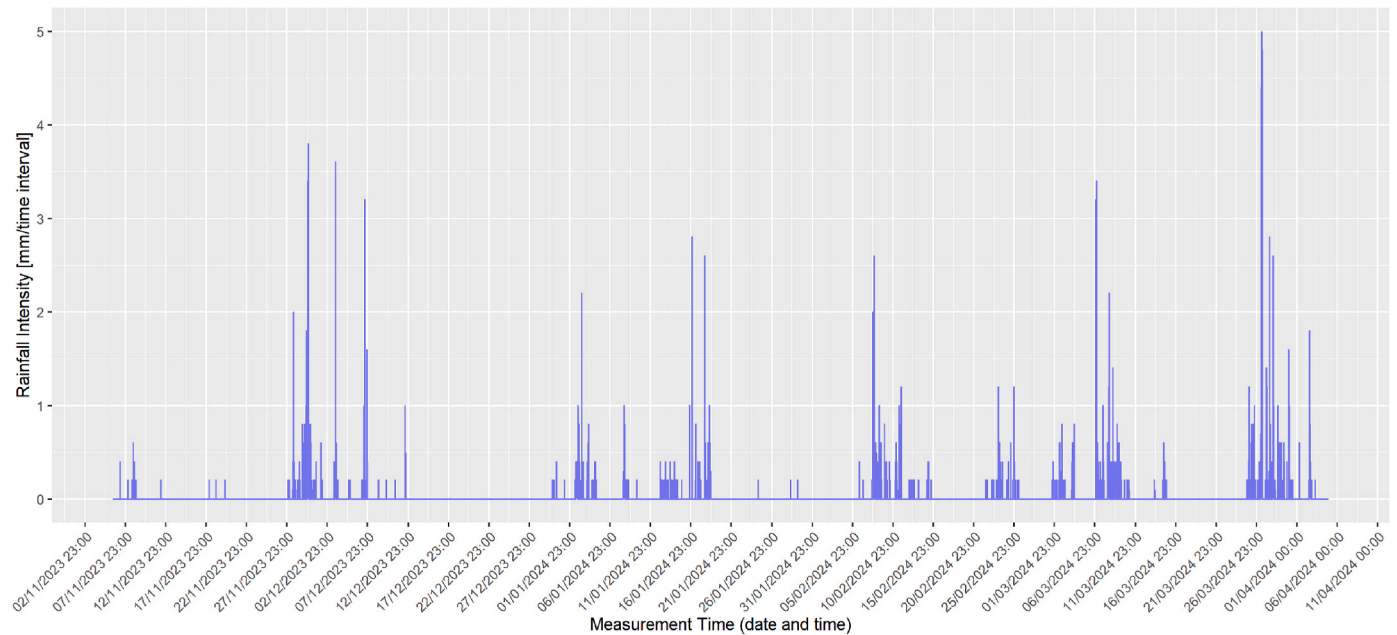


Fig. 6. Time series associated with tipping bucket gauge.



Fig. 7. Weighing gauge with storage capacity reached.

The analysis of the series of results presented in Figs. 5 and 6 took two steps: the first, aiming to evaluate the errors of the tipping bucket method, considering the weighing method as a reference, and the second, comparing the errors with the magnitude of the precipitation measured.

For the first purpose, six extreme events were selected, and the accumulated rainfall and the measurement error resulting from the two measurement methods were calculated, presented in Table 1. The weighing gauge was used as the reference equipment to estimate the relative measurement error of the tipping bucket method, P_e , given by the percentage difference (1) between the tipping bucket measurement, P_{tb} , and the weighing measurement, P_w .

Table 1

Accumulated rainfall and the measurement error for the rainfall events.

Periods	Weighing gauge Rainfall (mm)	Tipping bucket gauge Rainfall (mm)	Relative Measurement error of tipping bucket method (%)
November 28, 2023 to December 12, 2023	386,9	136,0	64,8
January 08, 2024 to January 19, 2024	155,1	89,2	42,5
February 06, 2024 to February 15, 2024	159,8	88,4	44,7
February 22, 2024 to February 26, 2024	34,5	20,6	40,3
March 01, 2024 to March 15, 2024	180,0	85,5	52,5
March 25, 2024 to April 03, 2023	496,8	162,0	67,4

$$P_e = 100 \cdot \frac{|P_{tb} - P_w|}{P_w} \quad (1)$$

The results obtained for accumulated rainfall for the periods considered show that the maximum accumulated rainfall occurred in the same period for both measurement methods.

For the periods considered, the measurement errors show a linear behaviour, indicating that the error increases are directly proportional to the rise of the rainfall intensity and the accumulated rainfall (Fig. 8 and Table 1).

The study adequately achieved the general relation exposed under the influence conditions. However, further analysis is needed to support this conclusion in more general terms, namely, by using more samples to increase the representativeness of the data for different types of influence conditions.

3.1. Analyses of high-intensity rainfall phenomena

Given the aim of this work, to analyse the data resulting from the two rainfall measurement methods, assessing the importance of having

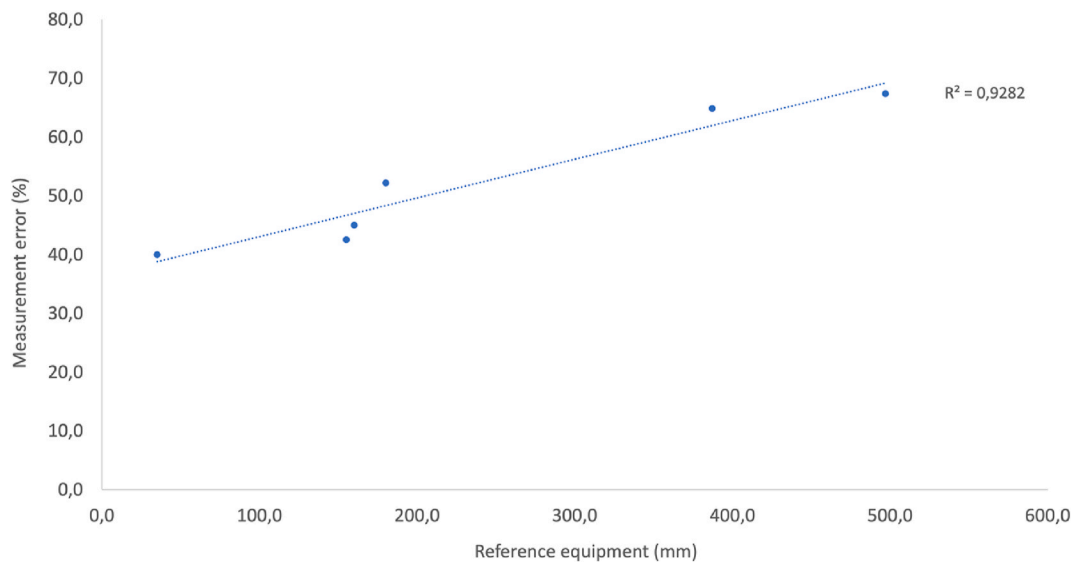


Fig. 8. Distribution of the relative measurement error of tipping bucket method the 6 events identified.

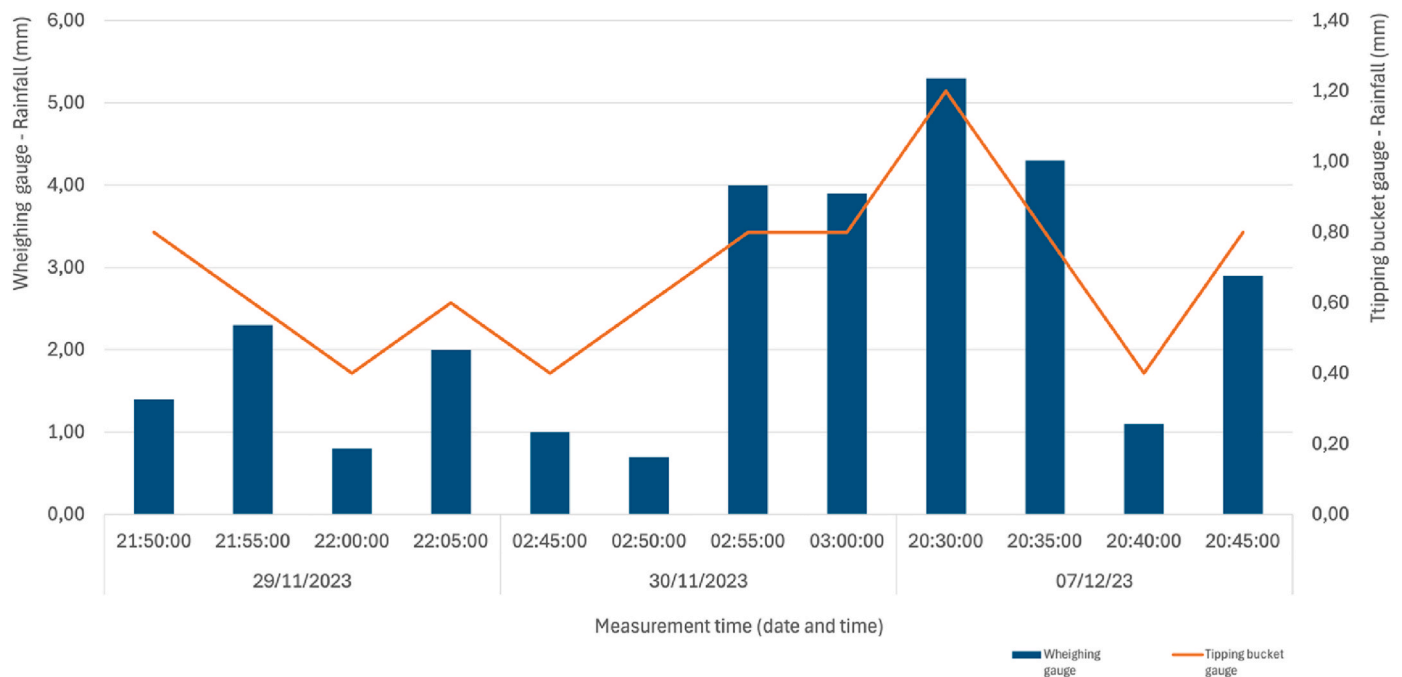


Fig. 9. Rainfall in the period November 29, 2023 to December 07, 2023.

rainfall data with higher data acquisition frequencies (10 seconds in the weighing gauge) and lower acquisition frequencies (5 minutes in the tipping bucket gauge) in the occurrence of high-intensity rainfall events, for the two periods identified in Table 1 with the highest accumulated rainfall, fifteen-minute intervals were selected to compare the accumulated rainfall obtained by the two rainfall measurement methods. Figs. 9 and 10 present the selected periods, being particularly noticed the differences obtained in several time intervals, .

To determine the relative measurement error of the tipping bucket method, applying (1), the weighing gauge was considered the reference equipment.

The values shown in Table 2 for the weighing gauge correspond to the accumulated rainfall over 5 minutes to compare the two measuring methods.

To assess the trend of measurement errors, Fig. 11 shows their

distribution. The results presented in Fig. 11 show a different behaviour to those presented in Fig. 8. These results show that the dispersion associated with the relative errors approaches the logarithmic function for a more significant number of observations. This approximation to the logarithmic function indicates that as the rainfall intensity increases, reflected in the accumulated rainfall, the error also increases, but not proportionally. This evidence may be related to the dynamic behaviour of other variables that affect the precipitation measurement process, namely the wind intensity and wind direction (which this study does not discuss in detail, being intended to be included in work under development).

3.2. Correlations analysis

The analysis was applied to the two rainfall time series with different

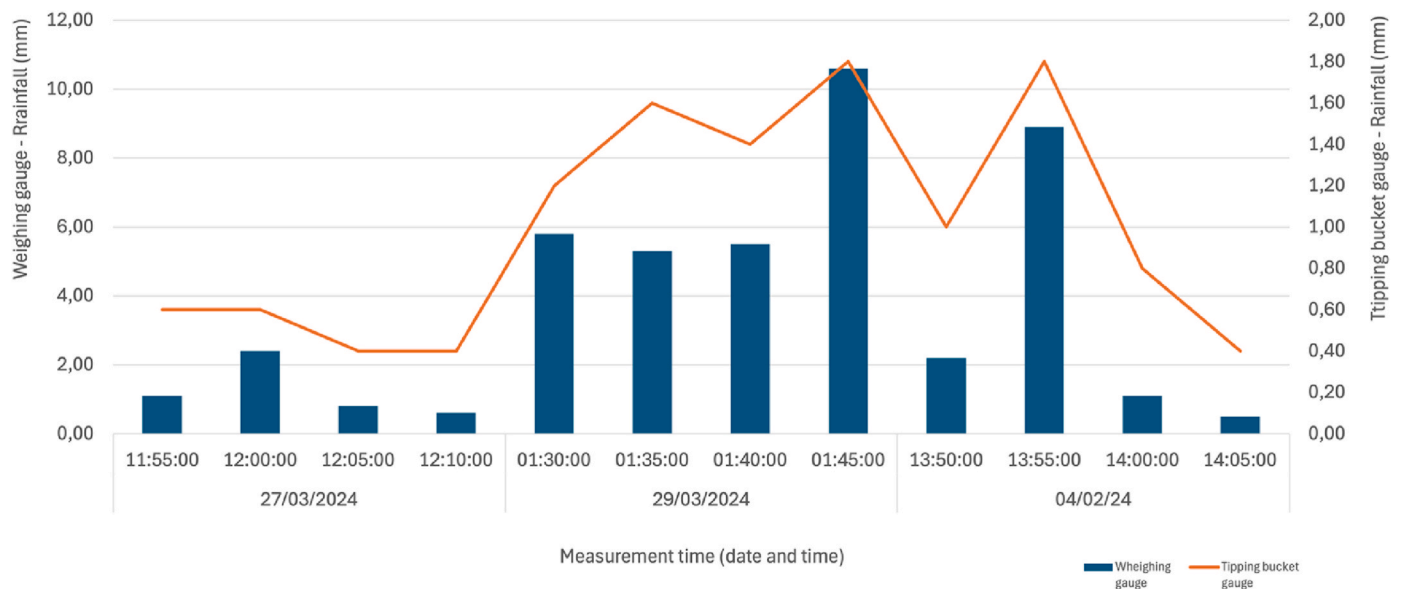


Fig. 10. Rainfall in the period March 27, 2024 to April 02, 2024.

Table 2

Measured rainfall and the relative measurement error of tipping bucket method for 15-min periods identified.

Periods	Measurement time		Wheighing gauge	Tipping bucket gauge	Error ^a (%)
	Date	Time	Rainfall (mm)	Rainfall (mm)	
November 28, 2023 to December 12, 2023	November 29, 2023	21:50:00	1,4	0,8	42,9
		21:55:00	2,3	0,6	73,9
		22:00:00	0,8	0,4	50,0
		22:05:00	2,0	0,6	70,0
	November 30, 2023	02:45:00	1,0	0,4	60,0
		02:50:00	0,7	0,6	14,3
		02:55:00	4,0	0,8	80,0
		03:00:00	3,9	0,8	79,5
	December 07, 2023	20:30:00	5,3	1,2	77,4
		20:35:00	4,3	0,8	81,4
20:40:00		1,1	0,4	63,6	
20:45:00		2,9	0,8	72,4	
March 25, 2024 to April 03, 2023	March 27, 2024	11:55:00	1,1	0,6	45,5
		12:00:00	2,4	0,6	75,0
		12:05:00	0,8	0,4	50,0
		12:10:00	0,6	0,4	33,3
	March 29, 2024	01:30:00	5,8	1,2	79,3
		01:35:00	5,3	1,6	69,8
		01:40:00	5,5	1,4	74,5
		01:45:00	10,6	1,8	83,0
	April 02, 2024	13:50:00	2,2	1,0	54,5
		13:55:00	8,9	1,8	79,8
14:00:00		1,1	0,8	27,3	
14:05:00		0,5	0,4	20,0	

^a Where it says 'Error,' it should read 'Relative Measurement error of tipping bucket method'. Due to space limitations, this designation was not included in the column.

acquisition frequencies. To carry out the study, the acquisition frequencies of the two series had to be synchronised, being used the functions of the RStudio © software.

The value obtained for the correlation of the two available data series, one associated with the weighing rainfall gauge and the other associated with the tipping bucket rainfall gauge, was 0.95, indicating a robust positive correlation between series. A correlation of 0.95 between precipitation series showing a good linear relationship (see also Figs. 5 and 6). The correlation obtained, bearing in mind that the two gauges

are installed in a short distance placement, supports the measurements' consistency. Fig. 12 shows the scatter plot, where it is clear that the greater the rainfall intensity, the greater the difference associated with the two measurement methods.

4. Conclusions

This study analysed two rainfall time series obtained by different measurement methods and with different acquisition frequencies. More accurate rainfall data makes it possible to detect periods of short duration, transitory variations in its behaviour, and this was evident in the results achieved, with differences in the rainfall intensity recorded by the two methods and reflected in the accumulated rainfall of more than 60 % for the same measurement periods but with different acquisition frequencies. Given the results obtained from the analysis, it is clear that as rainfall increases, the error associated with the tipping bucket gauge also increases, which can be justified not only by its measurement method but also by its acquisition frequency, i.e. the higher the defined acquisition frequency, the greater the likelihood of missing high rainfall events, especially those of short duration. Initially, it was found only for the events identified in the time series resulting from the two measurement methods that the associated errors showed a linear behaviour, i.e. as the rainfall intensity increases, the error of the measurements made by the weighing gauge increases proportionally.

However, with a more detailed analysis, for the six different days and with the 15 min. time intervals, it was found that the errors show a trend similar to the logarithmic function, i.e. as precipitation increases, the error increases, but not proportionally. However, the correlation between the two series showed an acceptable approximation of a linear correlation between the series. The effects of evaporation are directly related to the measurement of environmental variables (e.g. atmospheric temperature, relative humidity and wind intensity and direction). For this article, it was not possible to have data on these variables. Still, one solution adopted to minimise these effects, as mentioned in Ref. [11], was to reduce the frequency of acquisition between rainfall measurements, making it easier to detect the variability of the level of rainfall present in the bucket. Given the potential influence of environmental quantities (wind intensity and direction, atmospheric temperature, relative humidity) on rainfall measurement, these quantities are considered in proceeding studies. They could provide valuable information to be used later in the evaluation of measurement uncertainty analysis and for the correction of data applied in modelling and

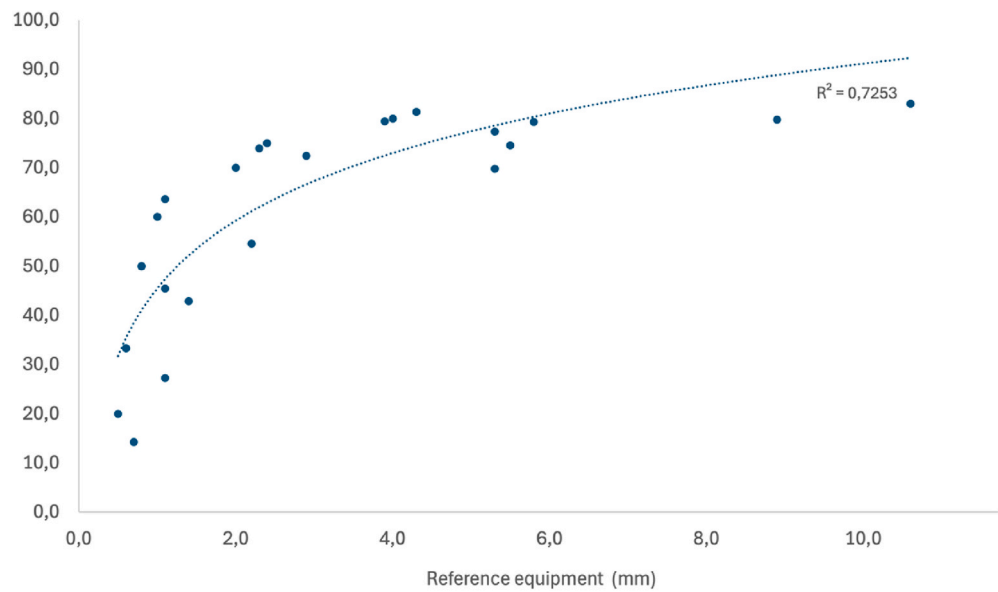


Fig. 11. Distribution of the relative measurement error of the tipping bucket method for the 15-min periods identified.

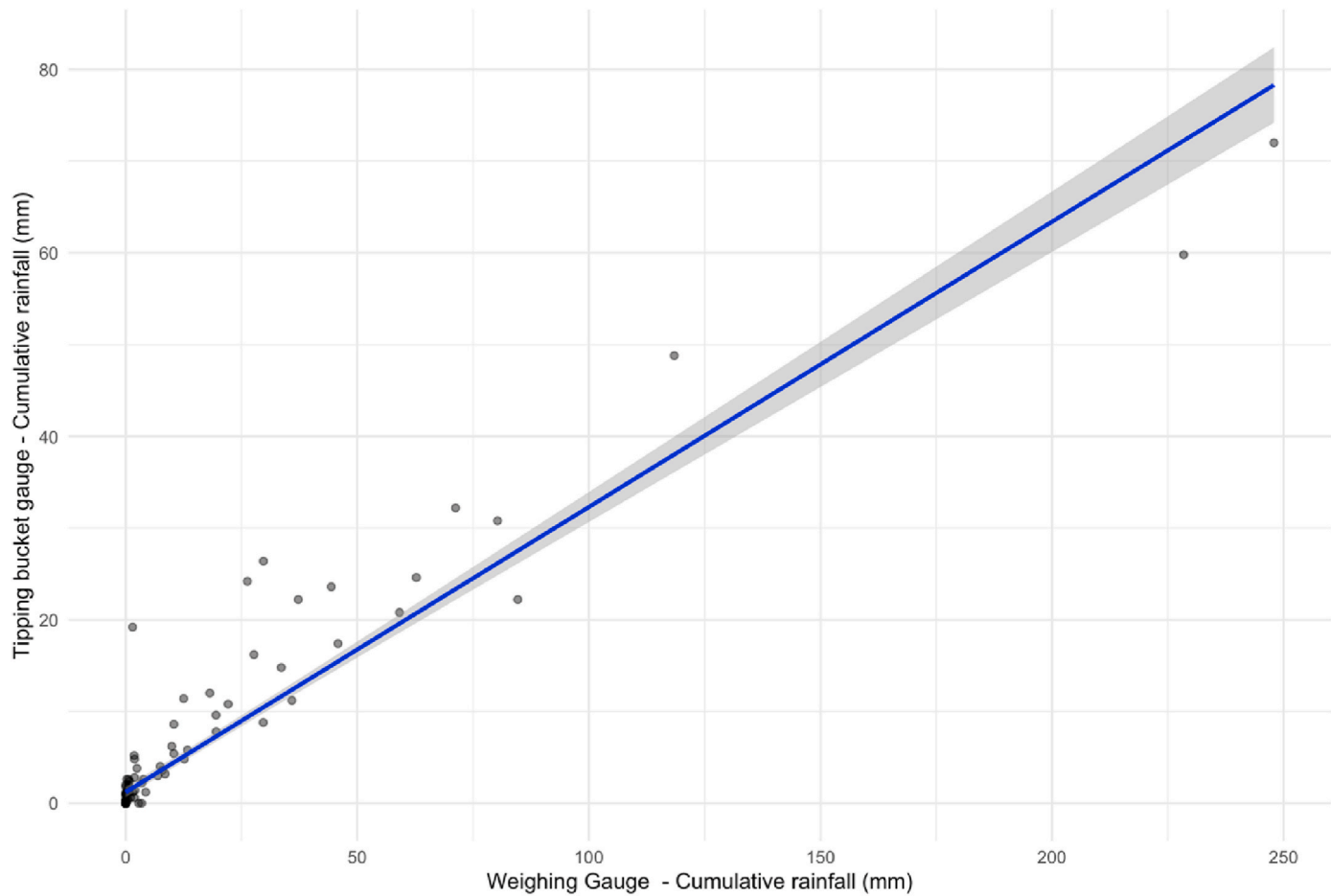


Fig. 12. Scatter plot associated with the two time series.

forecasting.

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Catarina Simões^{a,*}, Álvaro Ribeiro^a, Maria do Céu Almeida^a,
Dídia Covas^b, Gustavo Coelho^a, Alexandre Pinheiro^a

^a LNEC – National Laboratory for Civil Engineering, Lisbon, Portugal

^b CERIS, Instituto Superior Técnico, Universidade de Lisboa, Av. Rovisco
Pais 1, 1049-001, Lisboa, Portugal

* Corresponding author.

E-mail addresses: csimoes@lnec.pt (C. Simões), asribeiro@lnec.pt (Á. Ribeiro), mcalmeida@lnec.pt (M.C. Almeida), didia.covas@tecnico.ulisboa.pt (D. Covas), gfc Coelho@lnec.pt (G. Coelho), apinheiro@lnec.pt (A. Pinheiro).